Study of YOLO Dataset Configuration Files

In this week, I explored the dataset structure and meta configuration files used in YOLO models. These files are important because they define how training and validation data are organized for object detection tasks.

Typical YOLO Dataset Structure

```
dataset/
├── images/train
├── images/val
├── labels/train
├── labels/val
└── data.yaml
```

The images folder stores input images, while the labels folder stores annotation files for each image.

Purpose of YAML File

The YAML file contains key dataset information such as:

- base dataset path
- training image folder path
- validation image folder path
- class names
- number of classes

Example:

```
path: dataset
train: images/train
val: images/val

names:
0: person
1: bicycle
2: car
```

This means class 0 represents person, class 1 represents bicycle, etc.

Label Files

Each image has a matching text file.

Example:
image1.jpg
image1.txt

Contents: = 0 0.52 0.61 0.30 0.45

Format: class_id center_x center_y width height

These values are normalized between 0 and 1, which helps the model work with images of different sizes.

Importance

These files help YOLO understand:

- what objects are present
- object locations
- object classes
- which data is used for training/testing

References

- https://docs.ultralytics.com
- https://docs.ultralytics.com/datasets/detect/
- https://docs.python.org